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FA23-BCE-113

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Problem:

Use the method of steepest ascent to minimize:

$$z = (x_1 - \sqrt{5})^2 + (x_2 - \pi)^2 + 10$$

Solution:

first we transform it to maximization problem

$$z = -(x_1 - \sqrt{5})^2 - (x_2 - \pi)^2 - 10 \quad \text{— (A)}$$

We also need:

(a) starting point
(b) ending point

After sampling at random points, we find the starting point as under:

$$X_0 = [6.597, \ 5.891]^T$$

$$f(X_0) = -36.577$$

let the stopping condition be $\epsilon = 0.05$

It 1:

$$X_1 = X_0 + \lambda_1 \nabla f \big|_{x_0} \quad \text{— (1)}$$

now:

$$\nabla f = \begin{bmatrix} \partial f / \partial x_1 \\ \partial f / \partial x_2 \end{bmatrix} = \begin{bmatrix} -2(x_1 - \sqrt{5}) \\ -2(x_2 - \pi \end{bmatrix}$$

$$\nabla f|_{x_0} = \begin{bmatrix} -2(6.597 - \sqrt{5}) \\ -2(5.891 - \pi) \end{bmatrix} = \begin{bmatrix} -8.722 \\ -5.499 \end{bmatrix}$$

from (1):

$$x_1 = \begin{bmatrix} 6.597 \\ 5.891 \end{bmatrix} + \lambda_1 \begin{bmatrix} -8.722 \\ -5.499 \end{bmatrix}$$

&

$$\bar{X}_1 = \begin{bmatrix} 6.597 - \lambda_1 8.722 \\ 5.891 - \lambda_1 5.499 \end{bmatrix} \quad \text{— (2)}$$

Therefore:

$$f(\bar{X}_1) = -(6.597 - 8.722\lambda_1 - \sqrt{5})^2 - (5.891 - 5.499\lambda_1 - \pi)^2 - 10$$

$$f(\bar{X}_1) = -(4.361 - 8.722\lambda_1)^2 - (2.749 - 5.499\lambda_1) - 10$$

$$f(\bar{X}_1) = -106.312\lambda^2 + 106.307\lambda_1 - 36.575 \quad \text{— (3)}$$

$$f'(\bar{X}_1) = -2(1.6312)\lambda_1 + 106.307$$

$$f''(\bar{X}_1) = -2(1.6312) < 0$$

Therefore, $f(x_1)$ has global maximum at

$$f'(x_1) = 0$$

$$f'(\bar{x}_1) = -2(106.312)\lambda_1 + 106.307 = 0$$

$$\lambda_1 = \frac{106.307}{2(106.312)} = 0.5$$

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Using $\lambda_1 = 0.5$ in eq (2)

$$\bar{X} = \left[6.597 - (8.722)(0.5) \right]$$

We can obtain the same result by substituting $\lambda_1 = 0.5$ in eq (3)

$$f(\vec{x}_1) - f(\vec{x}_0) = -10.000 + 36.577$$

$$= 26.577 > \epsilon$$

It 2:

$$\vec{X}_2 = \vec{X}_1 + \lambda_2 \Delta f|_{x_1} \quad \text{— 5}$$

$$\vec{X}_1 = \begin{bmatrix} 2.236 \\ 3.142 \end{bmatrix}$$

$$\nabla f|_{x_1} = \begin{bmatrix} -2(2.236 - \sqrt{5}) \\ -2(3.142 - \pi) \end{bmatrix} = \begin{bmatrix} 0.0001 \\ -0.0008 \end{bmatrix}$$

Therefore:

$$\vec{X}_2 = \begin{bmatrix} 2.236 \\ 3.142 \end{bmatrix} + \lambda_2 \begin{bmatrix} 0.0001 \\ -0.0008 \end{bmatrix}$$

$$\vec{X}_2 = \begin{bmatrix} 2.236 + 0.0001\lambda_2 \\ 3.142 + (-0.0008)\lambda_2 \end{bmatrix} \quad \text{— (6)}$$

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$$f(\vec{x_2}) = -(2.36 + 0.0001\lambda_2 - \sqrt{5})^2 - (3.142 - 0.0002\lambda_2 - \pi)^2 - 10$$

$$f(\vec{x_2}) = -(0.000068 + 0.0001\lambda_2)^2 - (0.00041 - 0.0002\lambda_2) - 10$$

$$f(\vec{x_2}) = (-1\times10^{-6}\lambda_2^2 - 1.36\times10^{-8}\lambda_2 + 1\times10^{-8}) - (6.4\times10^{-7}\lambda_2^2 - 6.56\times10^{-7}\lambda_2 + 1.681\times10^{-7}) - 10$$

$$f(\vec{x_2}) = -6.5\times10^{-7}\lambda_2^2 + 6.696\times10^{-7}\lambda_2 - 10.0000001781$$

$$f(\vec{x_2}) = -2(6.7\times10^{-7}) < 0$$

Therefore $f(x_2)$ has global maxima at

$$f'(\vec{x_2}) = 0$$

$$\lambda_2 = \frac{6.696\times10^{-7}}{2(6.5\times10^{-7})} = 0.5151$$